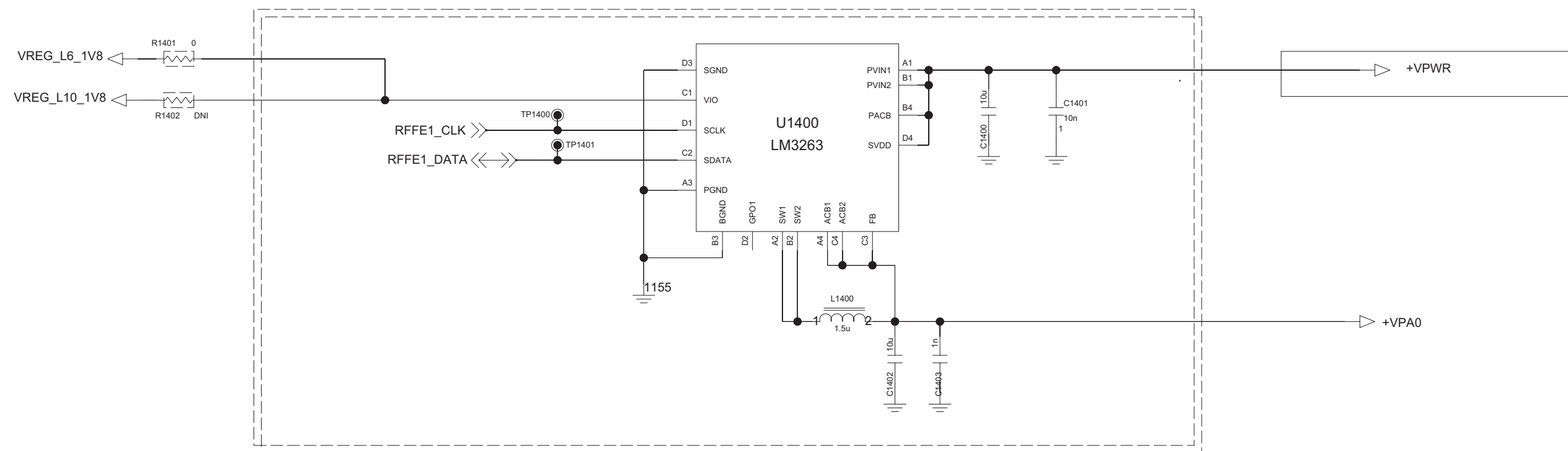
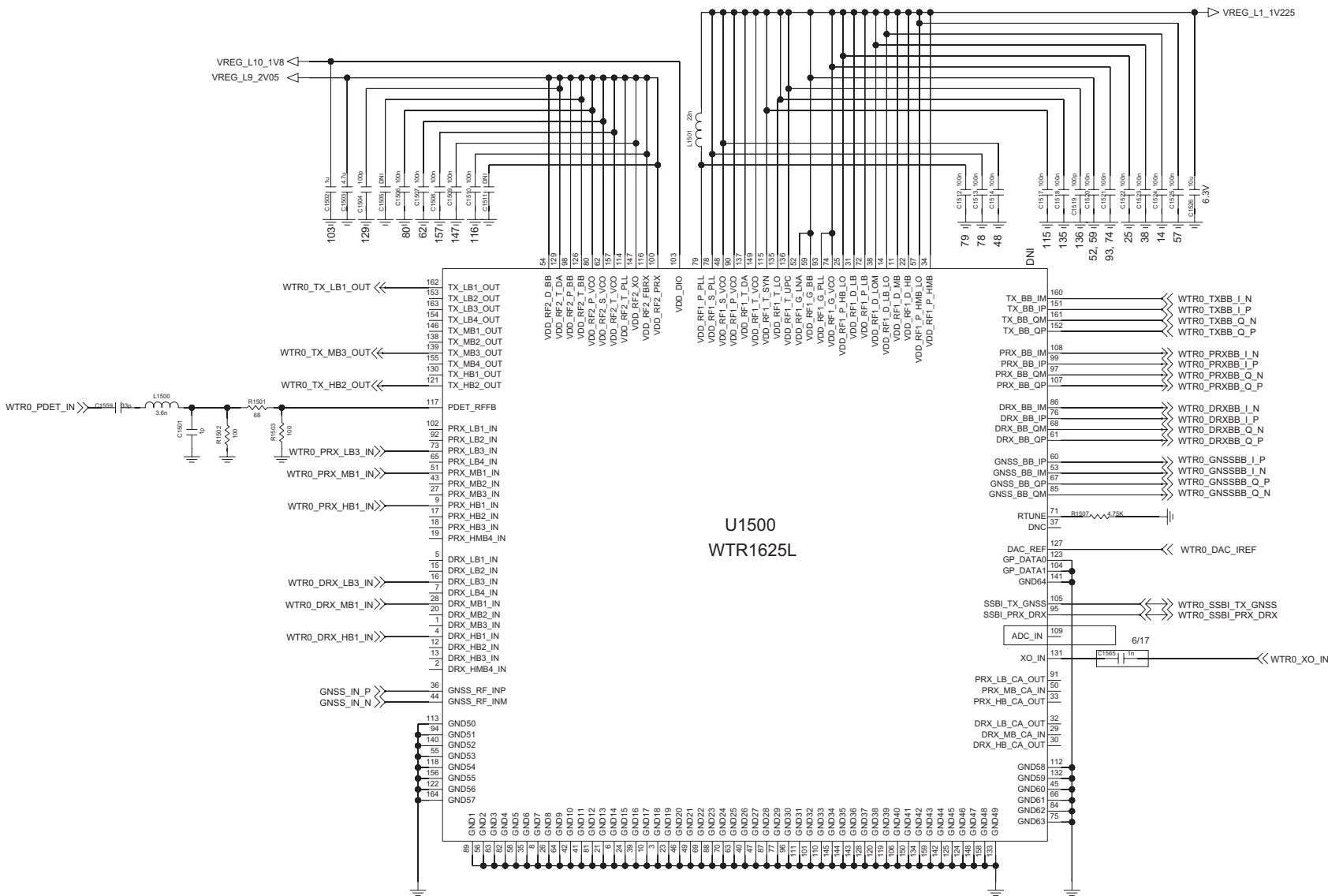


< 1-3-4-2_PT_APT_LM3263 > Rev_0.3

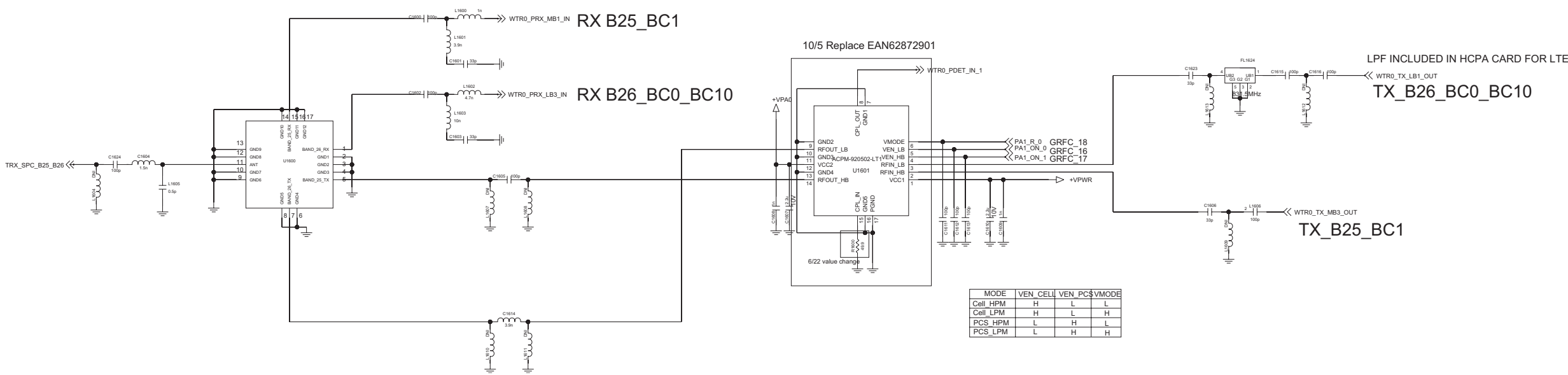


< 1-3-5-2_RFIC_SVLTE_WTR1625 > Rev_0.3

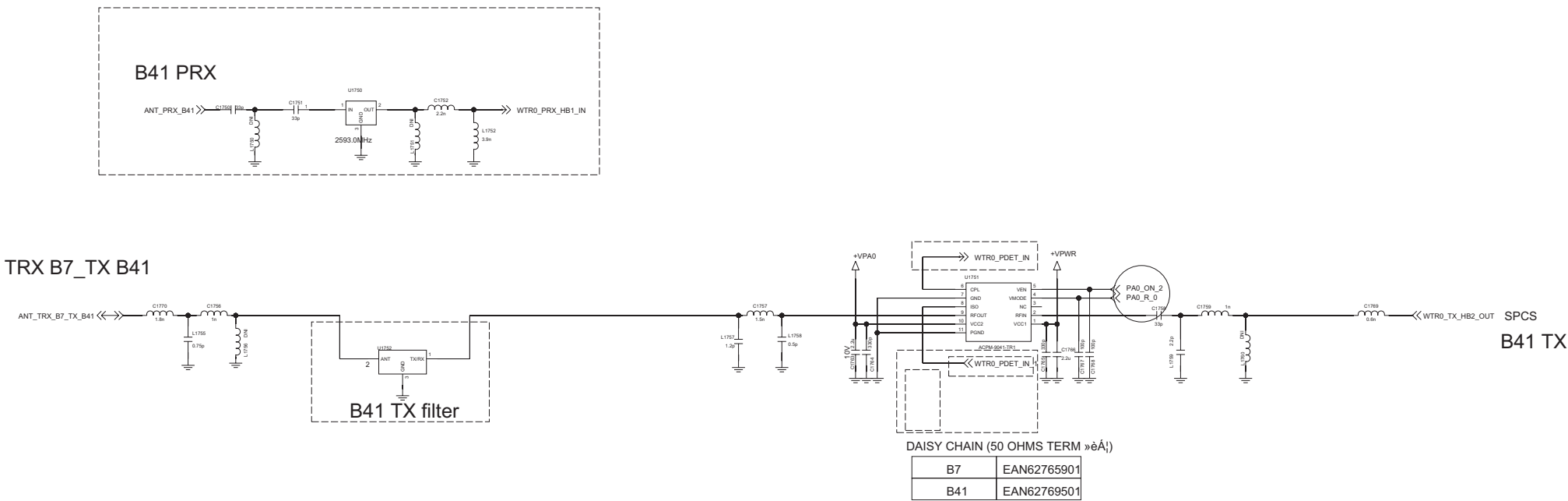


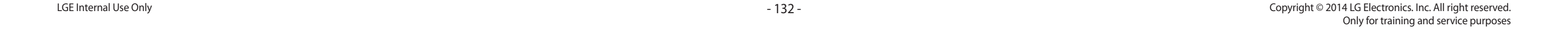
RF PORT MAP TABLE FOR WTR1625(L)+WFR2600 v01						
WTR1625 RF PORT MAPPING						
RF PORT	KR	DCM, KDDI	ATT, TMUS, SPC	EU, GLOBAL	VZW, USC	
TX, LB1, OUT	GSM, LB	GSM, LB	B5/26, BC0/10	B5/8, BC0	B5, B8	BC0
TX, LB2, OUT			GSM, LB	GSM, LB	GSM, LB	
TX, LB3, OUT	B17			B20		
TX, LB4, OUT	B5, B8	B6/18/19, BC0	B17		B12/13	
TX, MB1, OUT		B11/21		B34/39	B4	
TX, MB2, OUT	GSM, HB	GSM, HB	GSM, HB	GSM, HB	GSM, HB	
TX, MB3, OUT	B1, B2	B1, BC6	B1/2/25, BC1	B1/2	B1, B2	BC1
TX, MB4, OUT	B3, B4		B3/4	B3		
TX, HB1, OUT	B7	B41	B7	B7		
TX, HB2, OUT		B41	B41	B38/40		
PRX, LB1, IN	850, B5	B6/18/19, BC0	B29	B20		
PRX, LB2, IN	900, B8	900	900	900, B8	B8, G900	
PRX, LB3, IN			850, B5/26, BC0/10	850, B5, BC0	B5, G850	BC0
PRX, LB4, IN	B17		B17		B12/13	
PRX, MB1, IN	B2	B11/21	B2/25, BC1	B2	B4	
PRX, MB2, IN	GSM, HB	GSM, HB	GSM, HB	GSM, HB	G1800	
PRX, MB3, IN			B3	B3	B2, G1900	BC1
PRX, HB1, IN			B41	B38/40		
PRX, HB2, IN			B41	B34/39		
PRX, HB3, IN	B7		B7	B7		
PRX, HMB4, IN		B1, BC6	B1/4	B1	B1	
DRX, LB1, IN	B8		B29	B20		
DRX, LB2, IN	B5	B18/19, BC0		B8		
DRX, LB3, IN			B5/26		B5	BC0
DRX, LB4, IN	B17		B17		B12/13	
DRX, MB1, IN		B11/21	B2/25, BC1		B4	
DRX, MB2, IN				B39		
DRX, MB3, IN			B3	B3	B2	BC1
DRX, HB1, IN			B41	B38		
DRX, HB2, IN				B40		
DRX, HB3, IN	B7		B7	B7		
DRX, HMB4, IN		B1, BC6	B1/4	B1		
WFR1620 RF PORT MAPPING						
RF PORT	KR	DCM, KDDI	ATT, TMUS, SPC	EU, GLOBAL		
PRX, MB1, IN			B2/25, BC1			
PRX, MB2, IN	B3		B3			
PRX, MB3, IN	B1/4		B1/4			
DRX, MB1, IN			B2/25, BC1			
DRX, MB2, IN	B3					
DRX, MB3, IN	B1/4		B1/4			

< 1-3-6-2_TRX_SPCS_BC0_BC1 > Rev_0.1



< 1-3-6-7_TRX_B7_TX_B41 > Rev_0.1

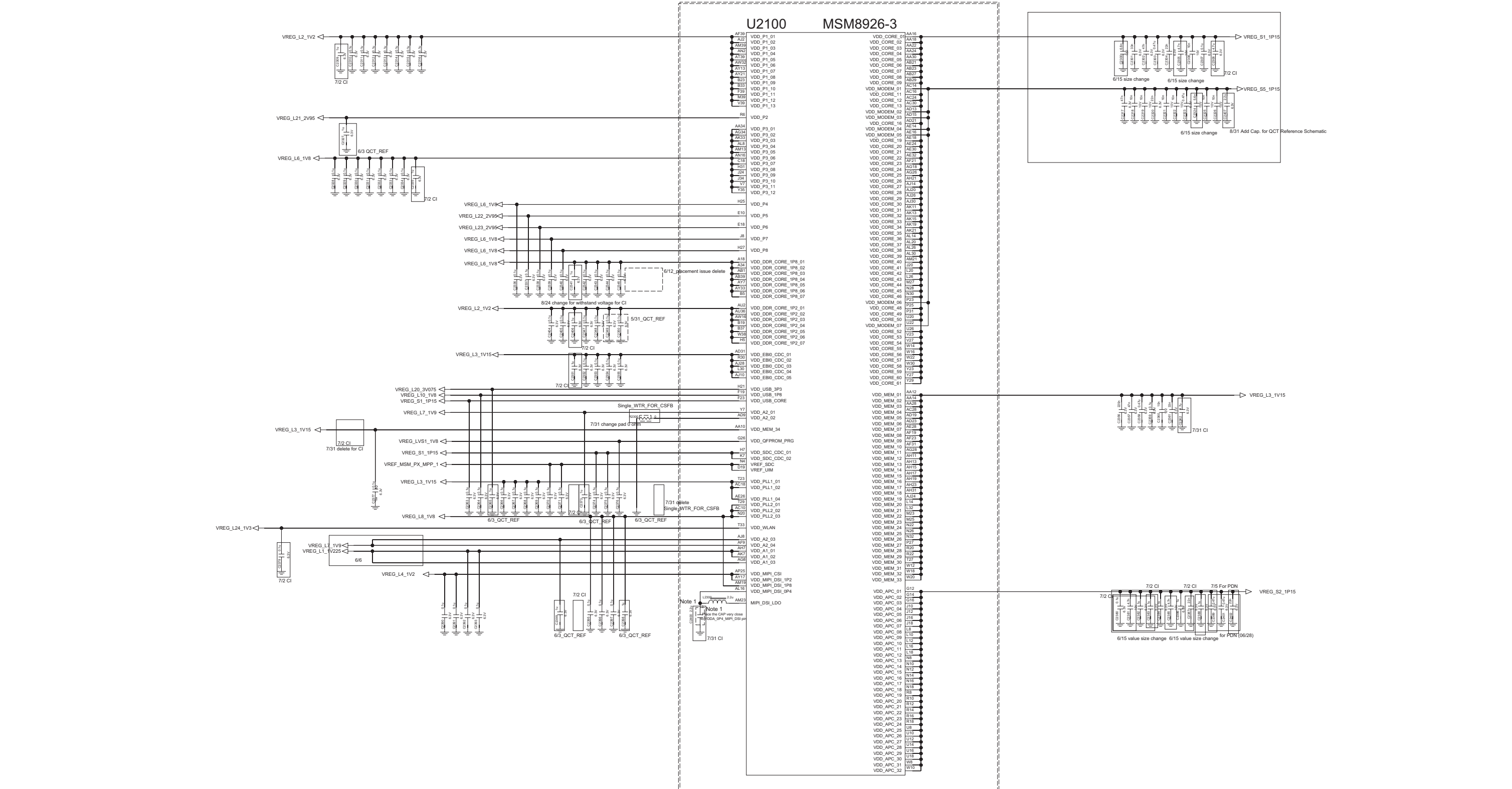




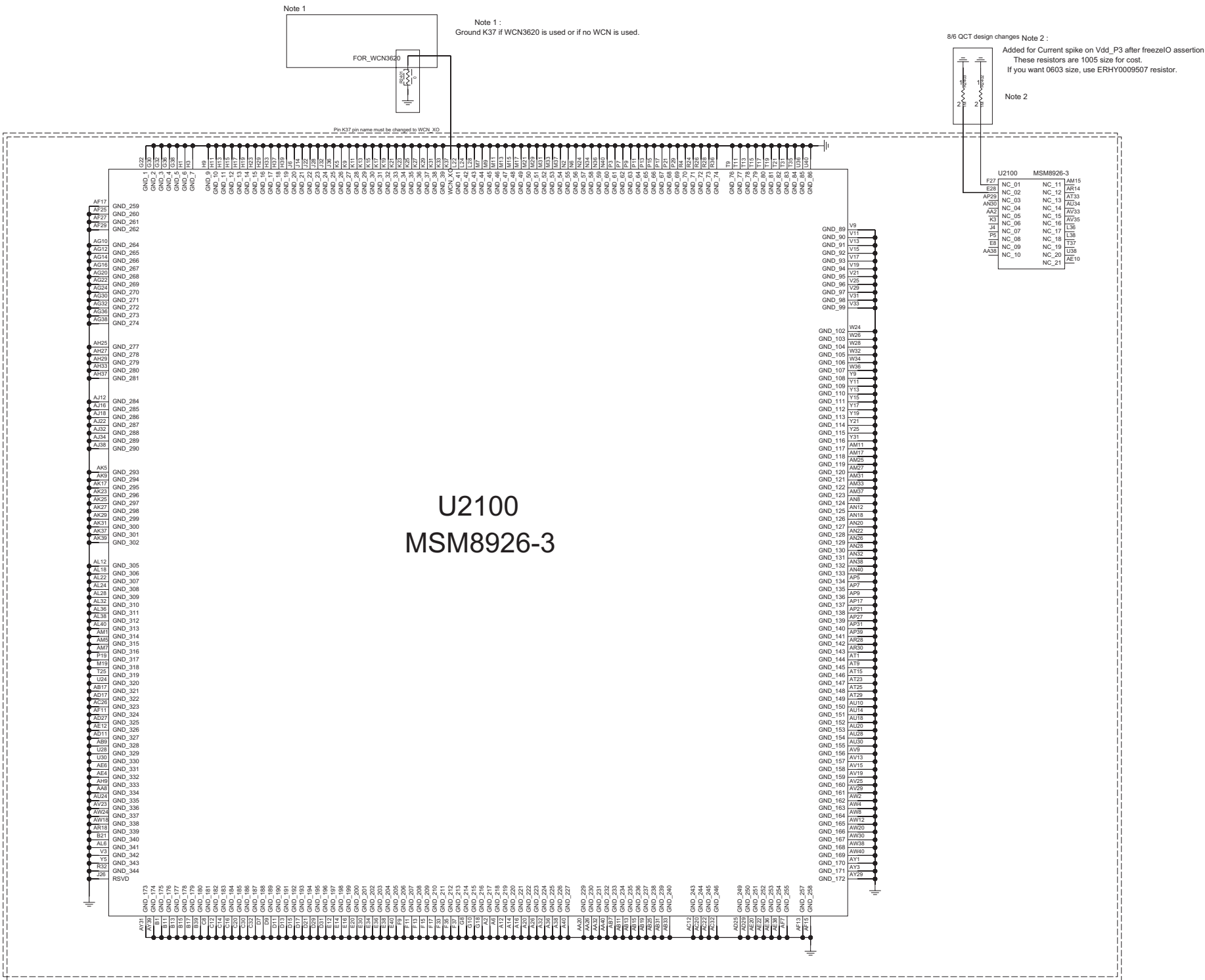


Note 8

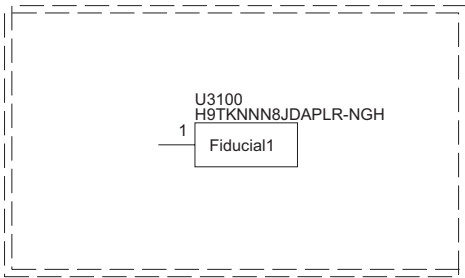
Note 8
If you don't use TDMB, delete this box



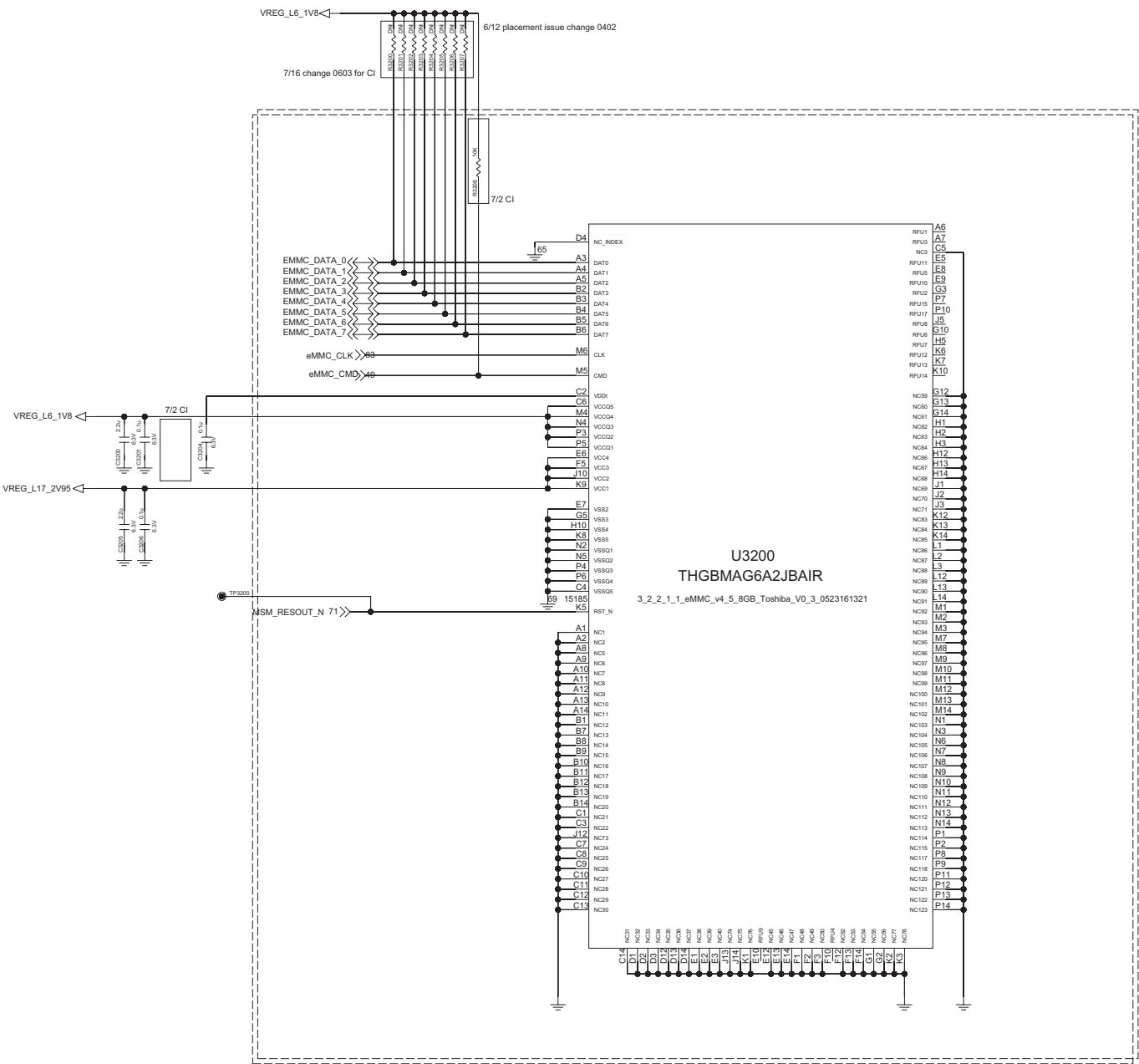
<2-1-3-5-4_MSM8926-3_GND_NC> Rev_0.3

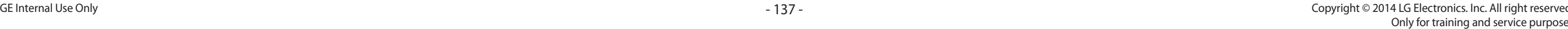


<3-1-2-3-3_LPDDR2_POP_8Gbit_Micron> Rev_0.3

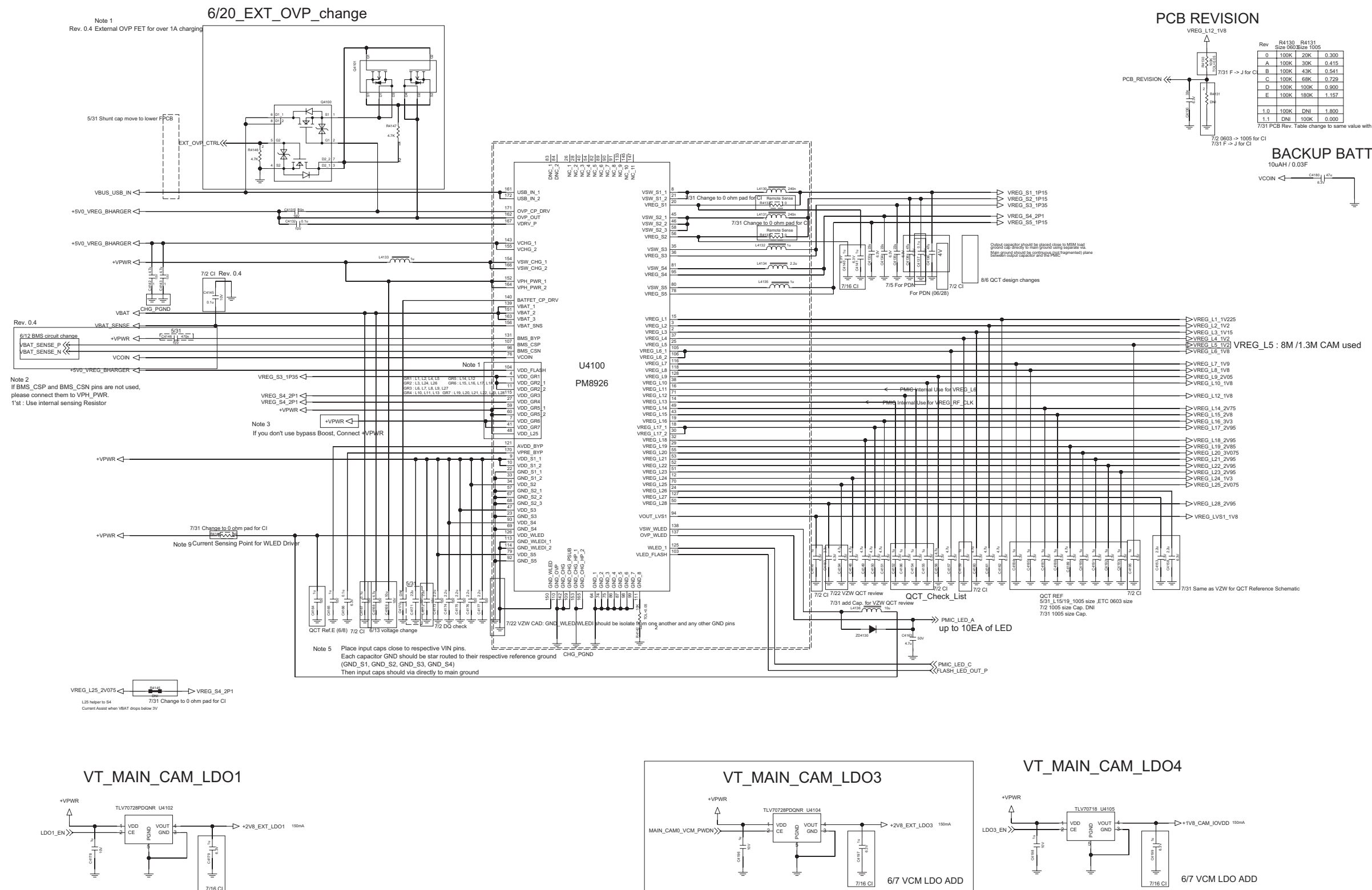


< 3-2-2-1-1_eMMC_v4_5_8GB_Toshiba > Rev_0.3

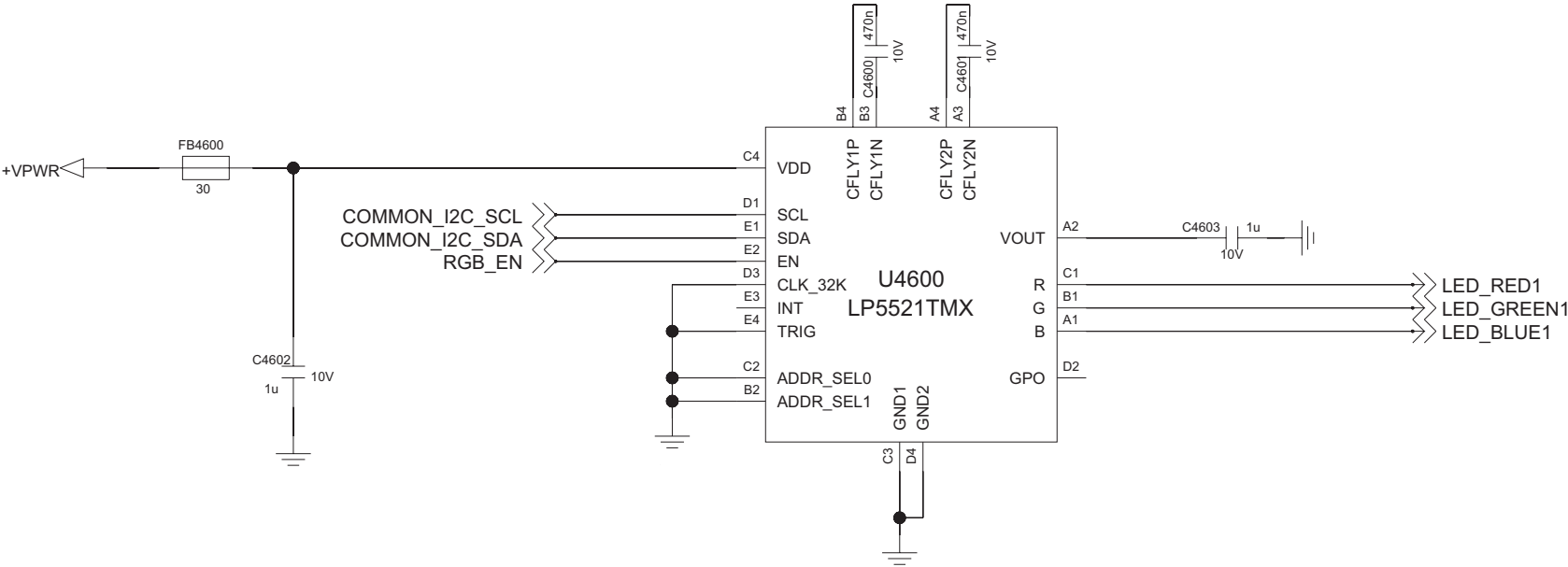




<4-1-3-2_PMIC_PM8926_POWER> Rev_0.6

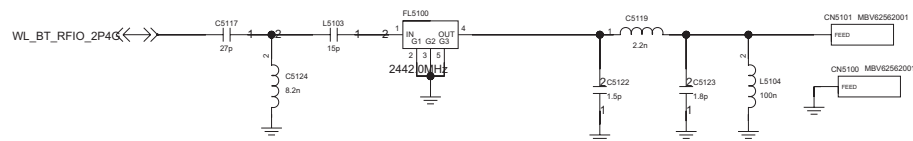


<4-6-1_LED_Driver_IC_LP5521> REV.0.4

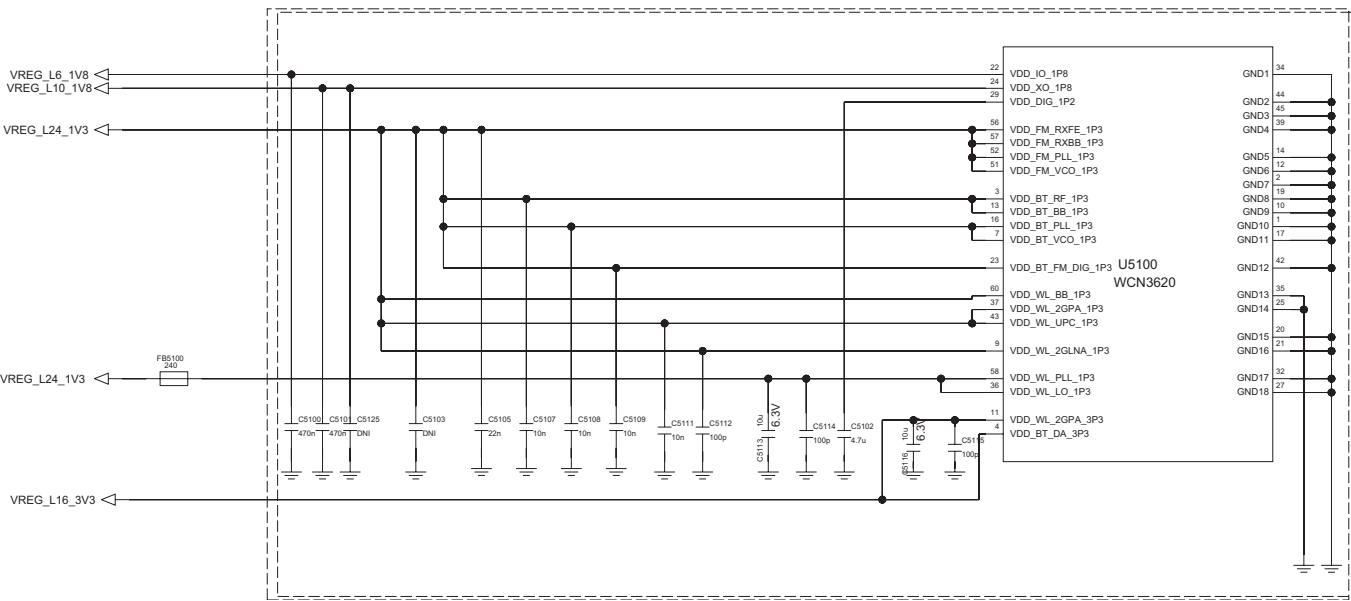


<5-1-1-4> BT/WiFi Combo_WCN3620 REV.0.4

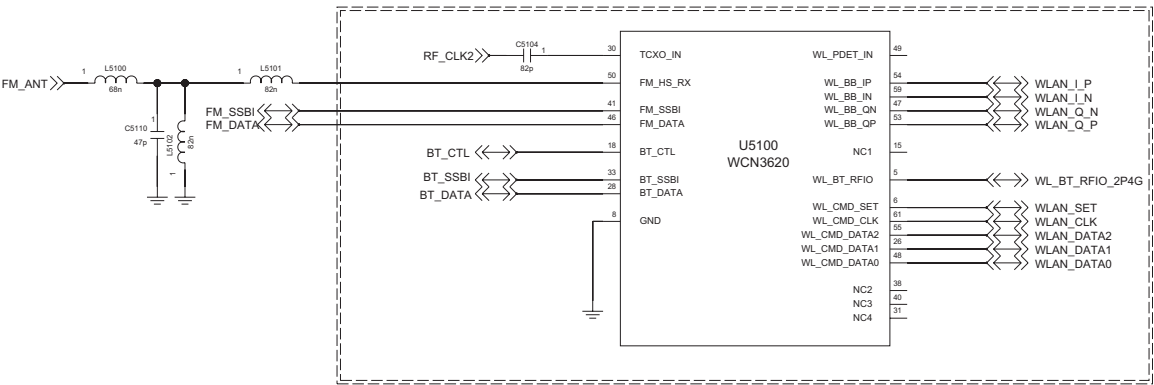
RF Filter

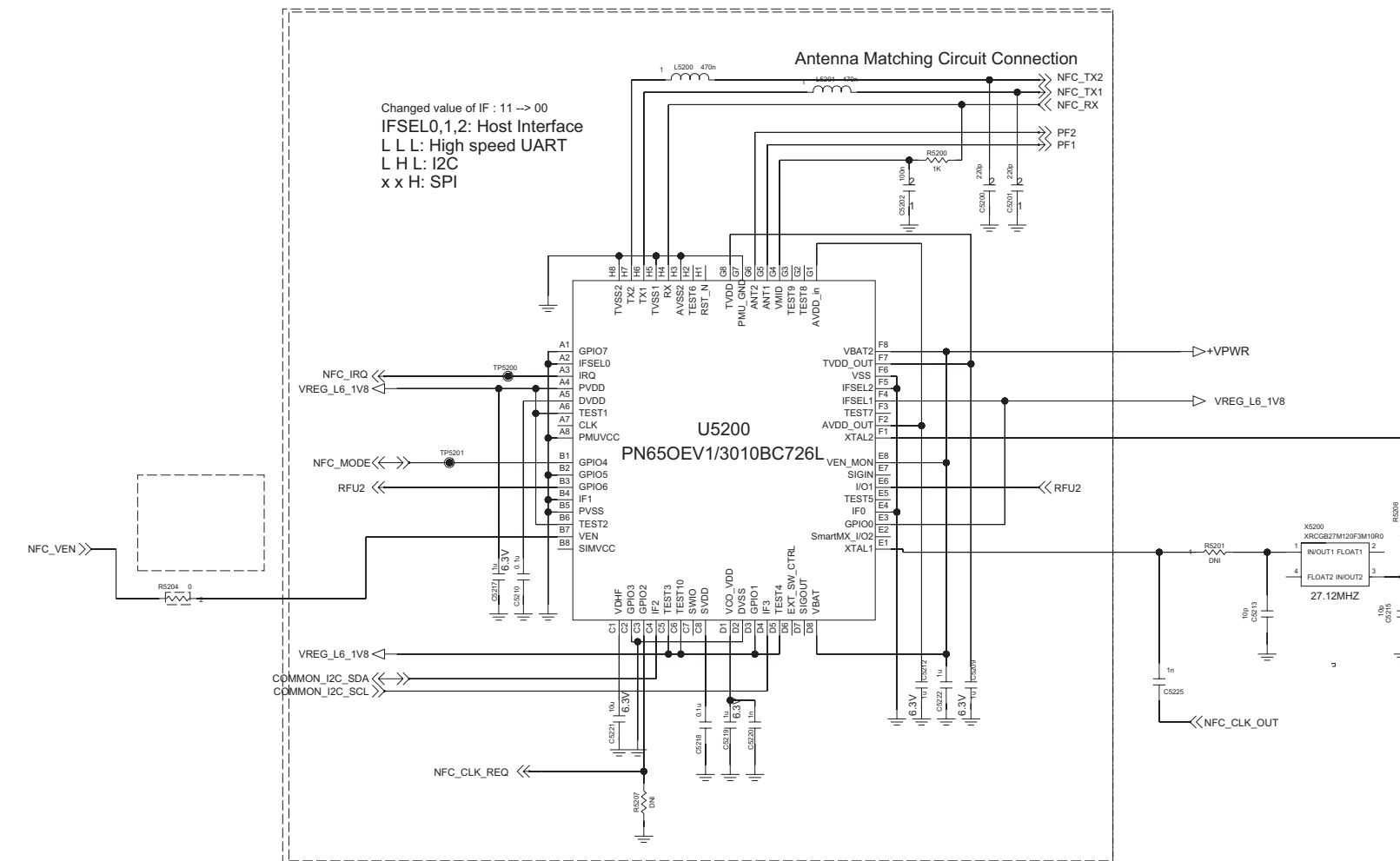


POWER/GND

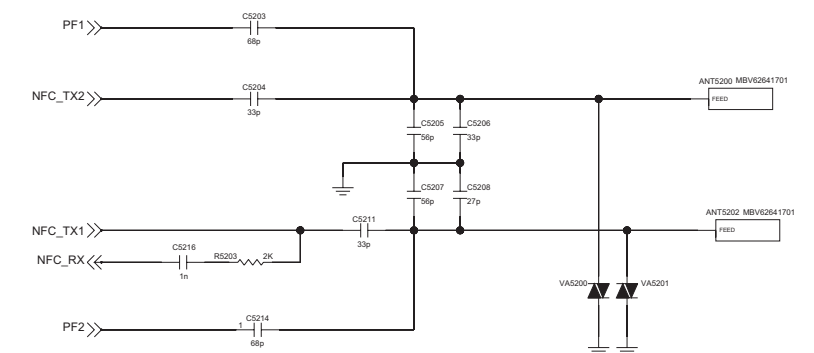


RF/CONTROL

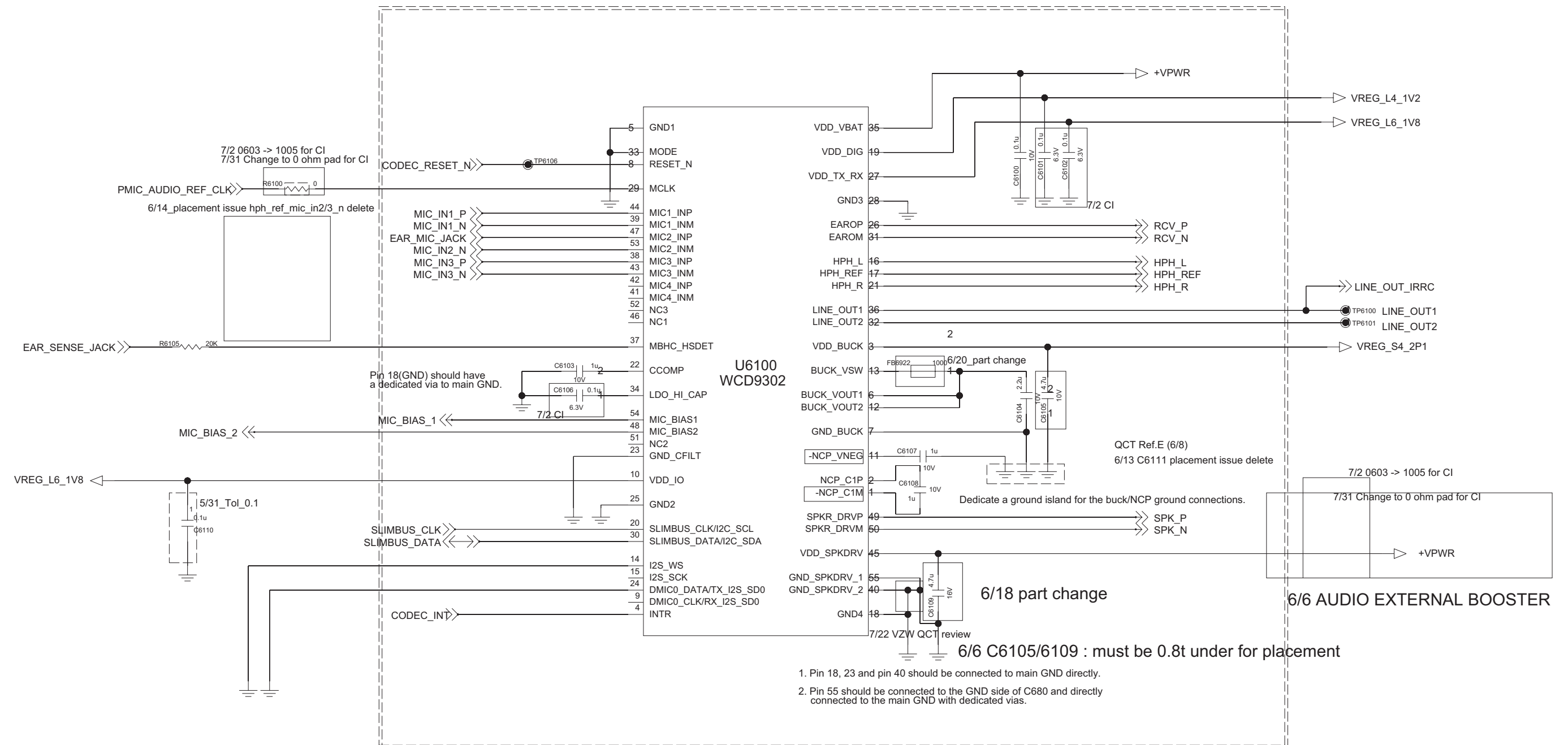




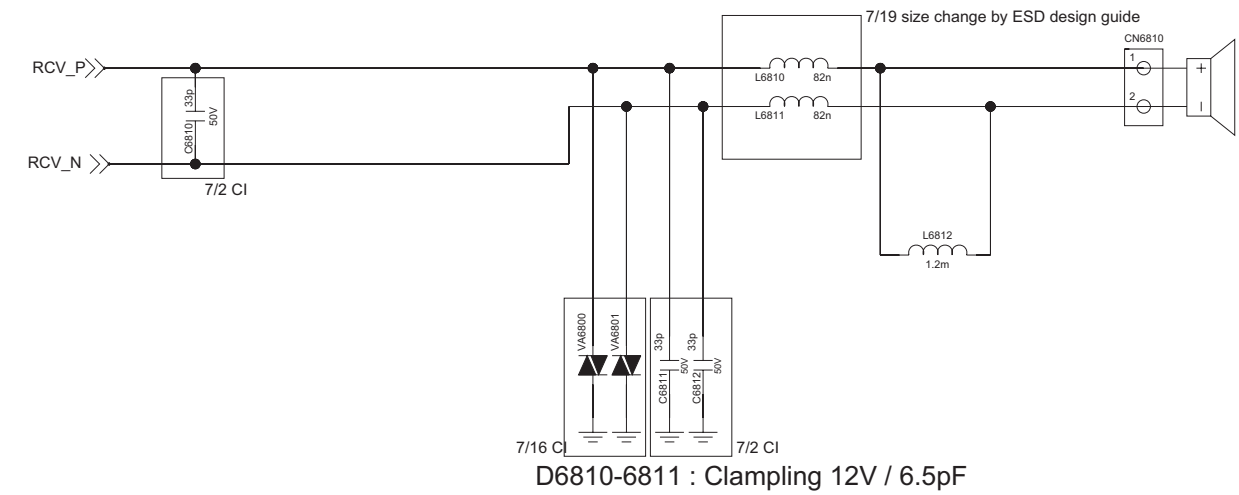
NFC Antenna



< 6-1-1-1_Codec_WCD9302 > Rev_0.3



< 6-8-1_Receiver_Only > Rev_0.4

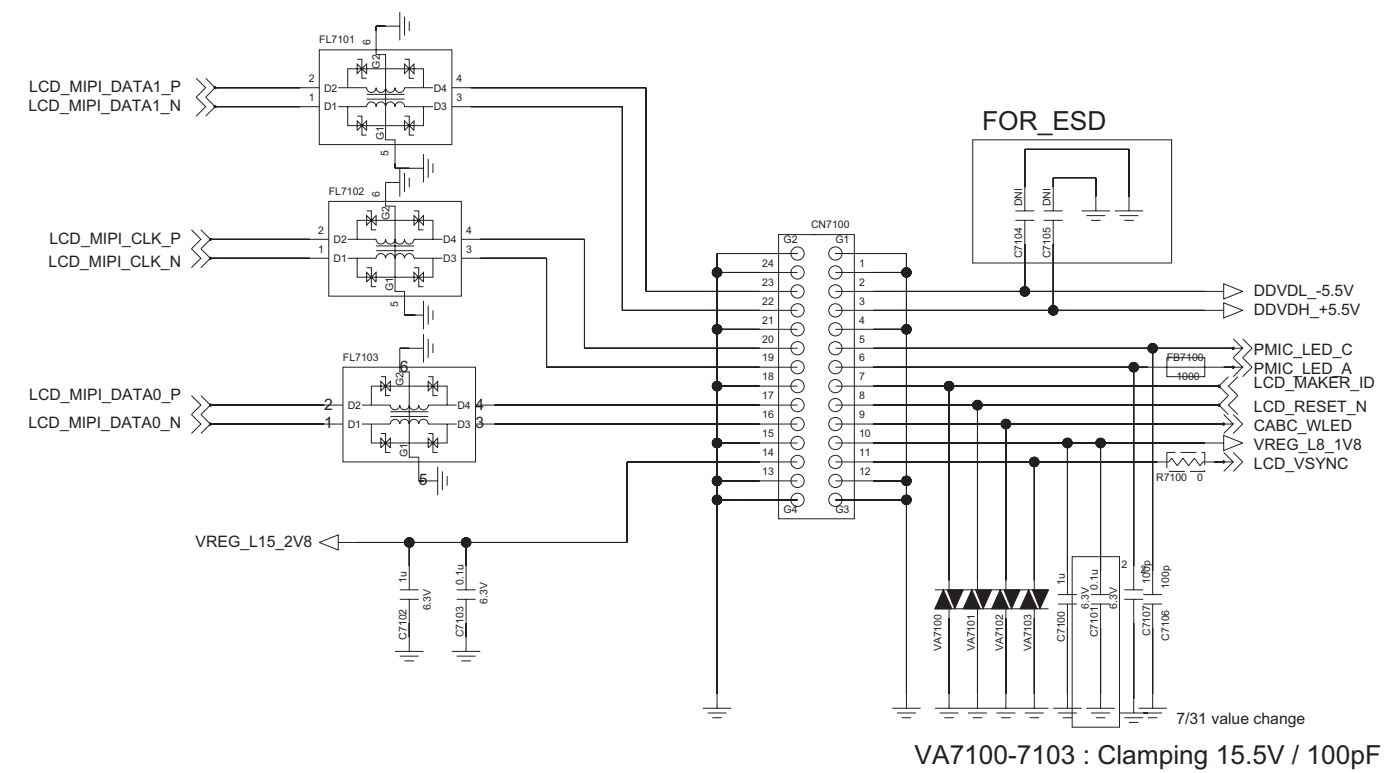


SUB_MIC_LOWER_FPCB



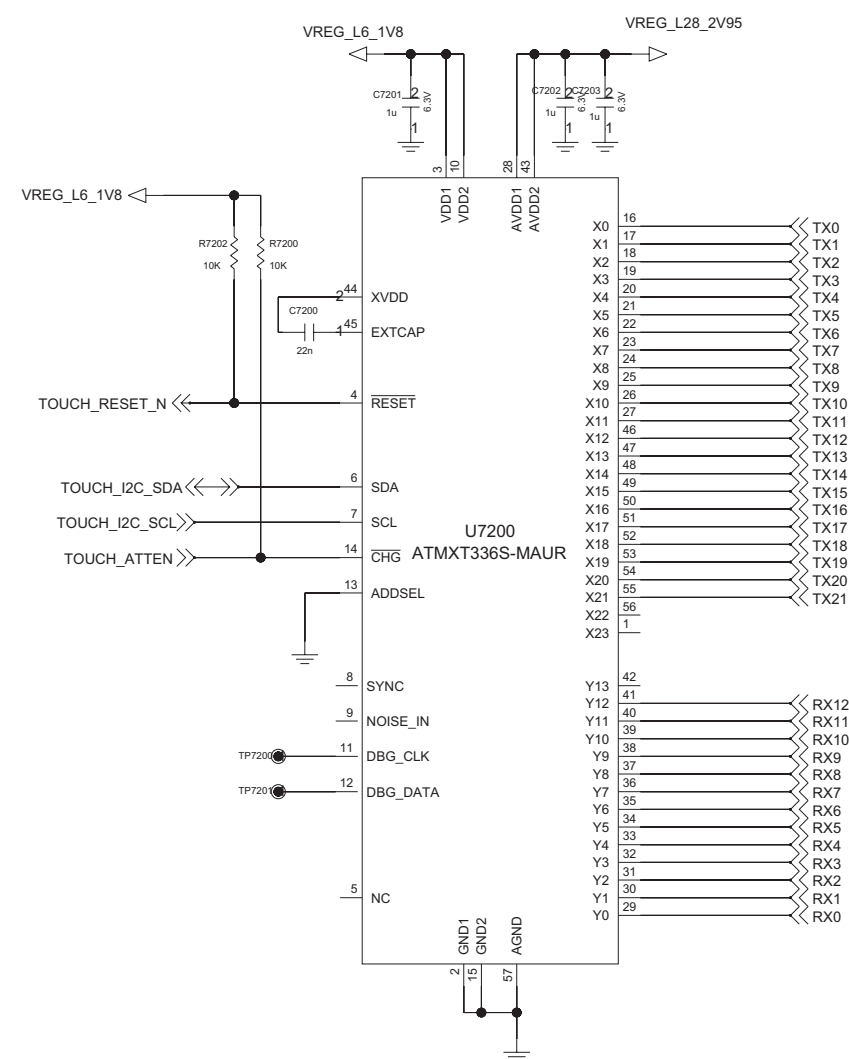
< 7-1-6-2-1_4.7inch_qHD > Rev_0.4

4.7" / qHD / LM468VN1A / P760

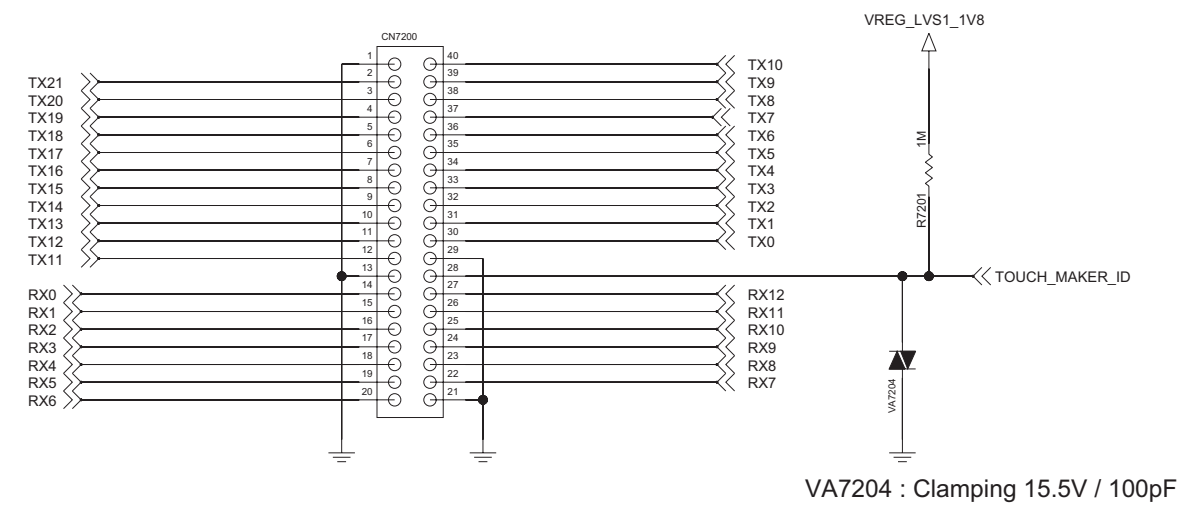


< 7-2-2-1_MMS-136_Touch_CoB_Atmel_mXT336S > Rev_0.3

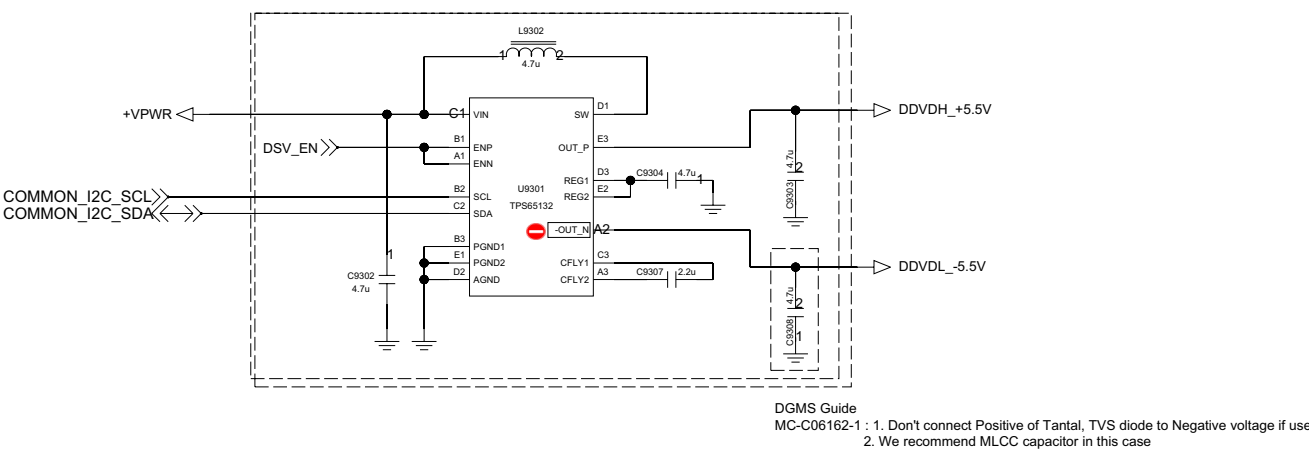
TOUCH COB



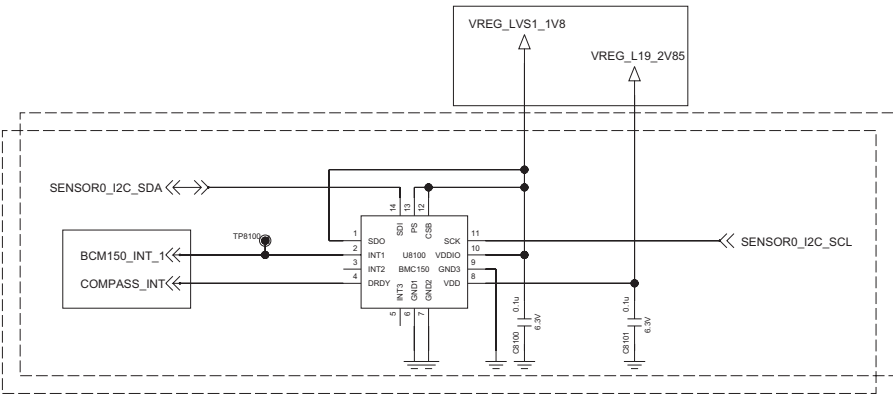
Touch Connector
(B to B type)



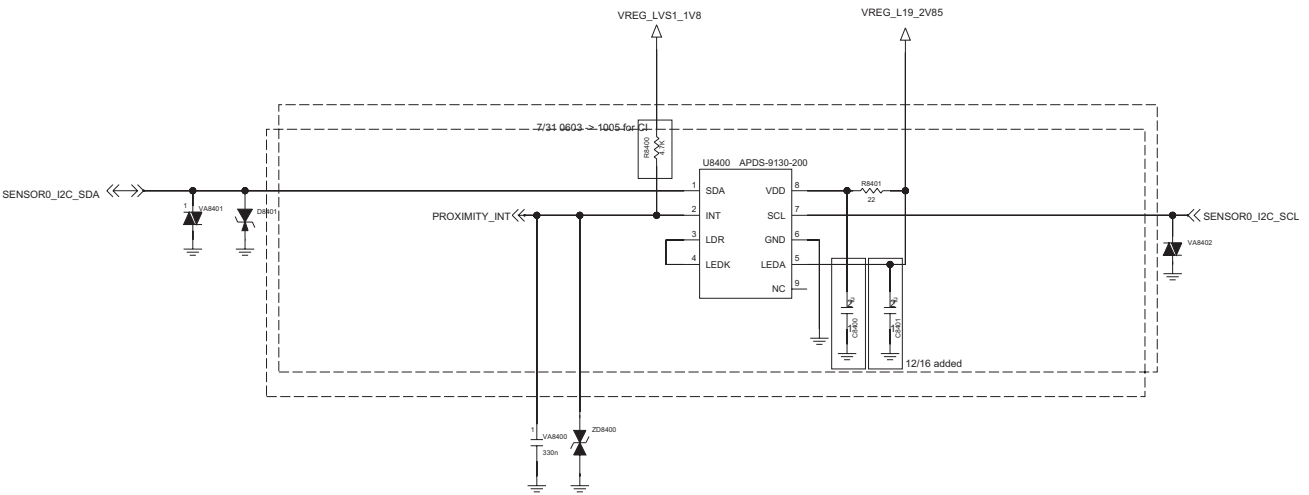
<7-5-1_DSV_TPS65132>Rev_0.3



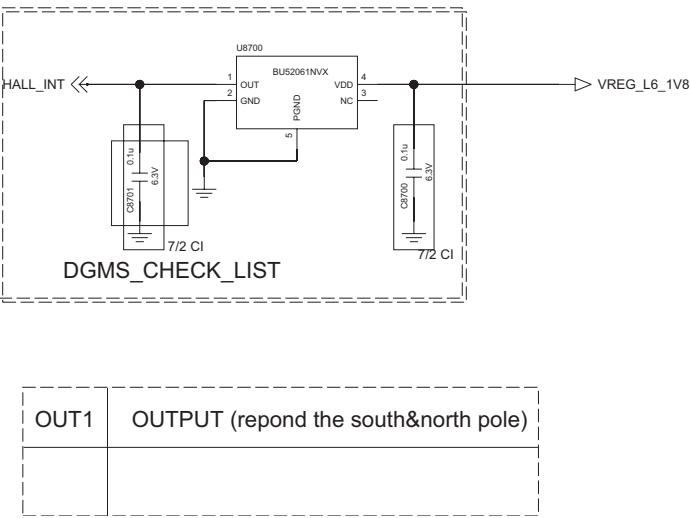
<8-1-2-1_Accel_Compass_BMC150> REV.0.4



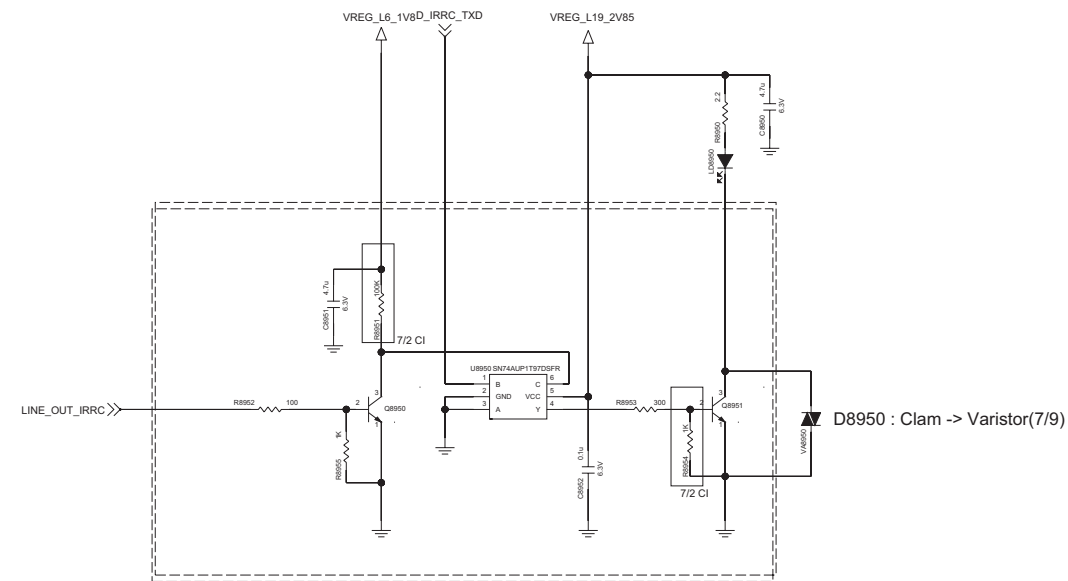
<8-4-1-1_Proximity_APDS-9130> REV.0.3



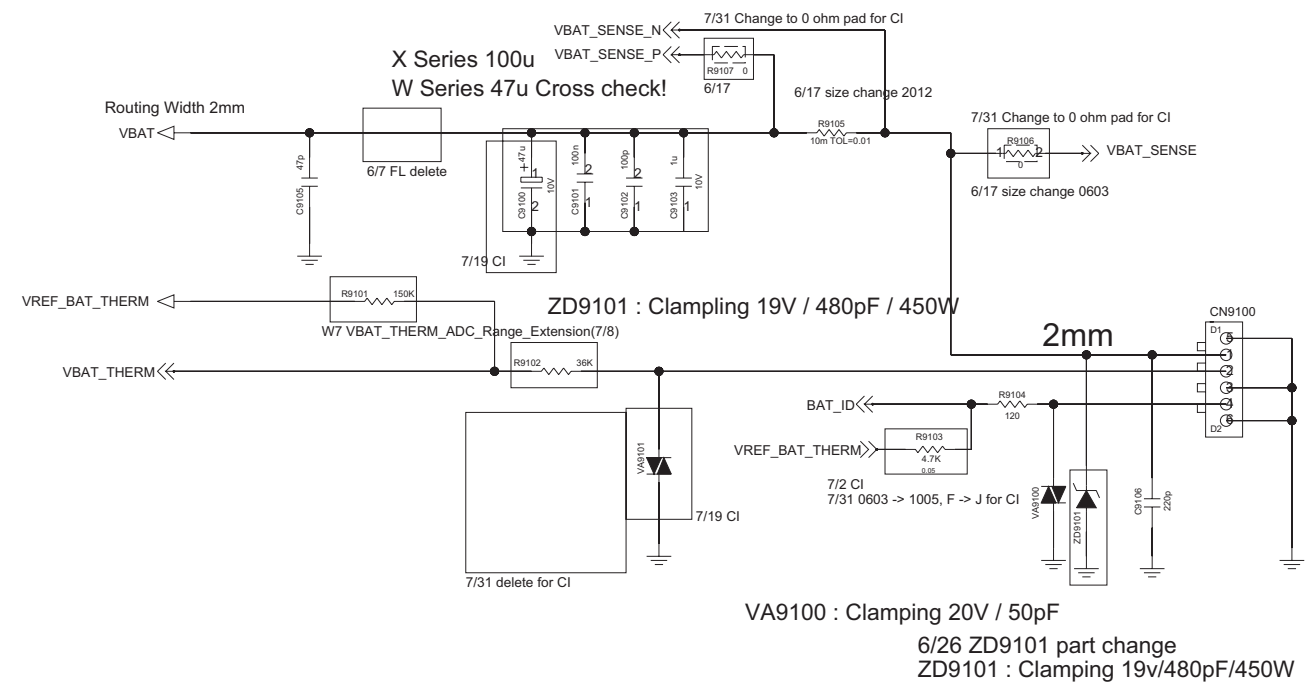
<8-7-1-1_Hall IC_BU52061NVX>Rev_0.3



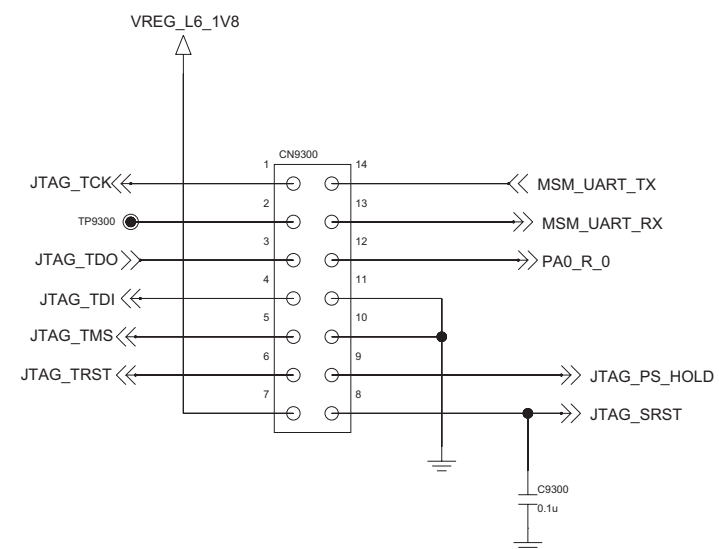
< 8-9-1-2_IRRC_w/o_driver_IC > Rev_0.4



<9-1-2_Battery_CNT_4P>REV.0.4

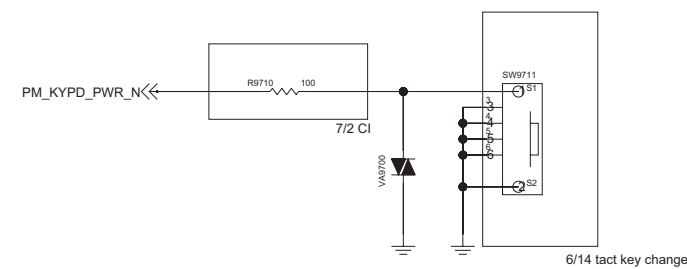


< 9-3-1_JTAG > Rev_0.4

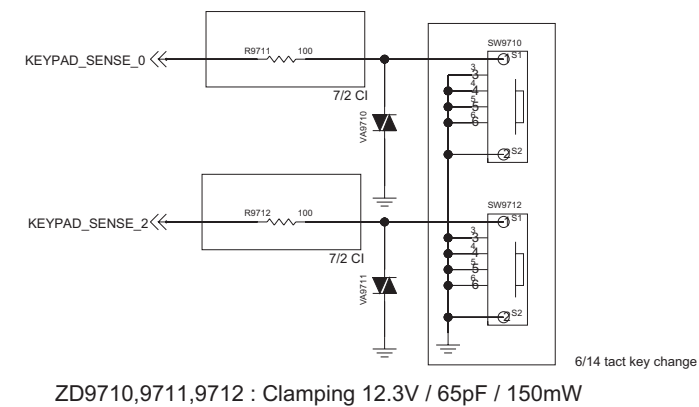


< 9-7-1_Power_Lock_Volume_Key > Rev_0.4

POWER & LOCK KEY

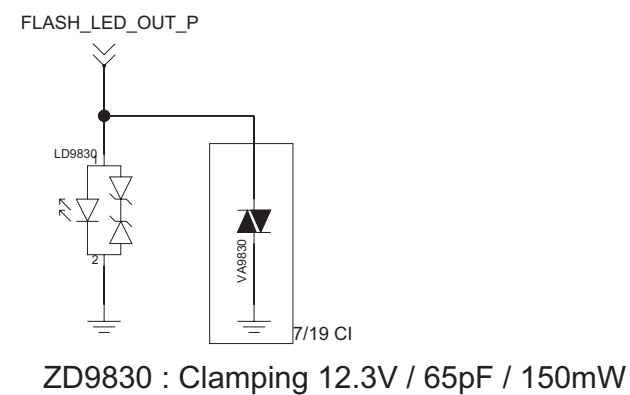


Vol. UP/DOWN



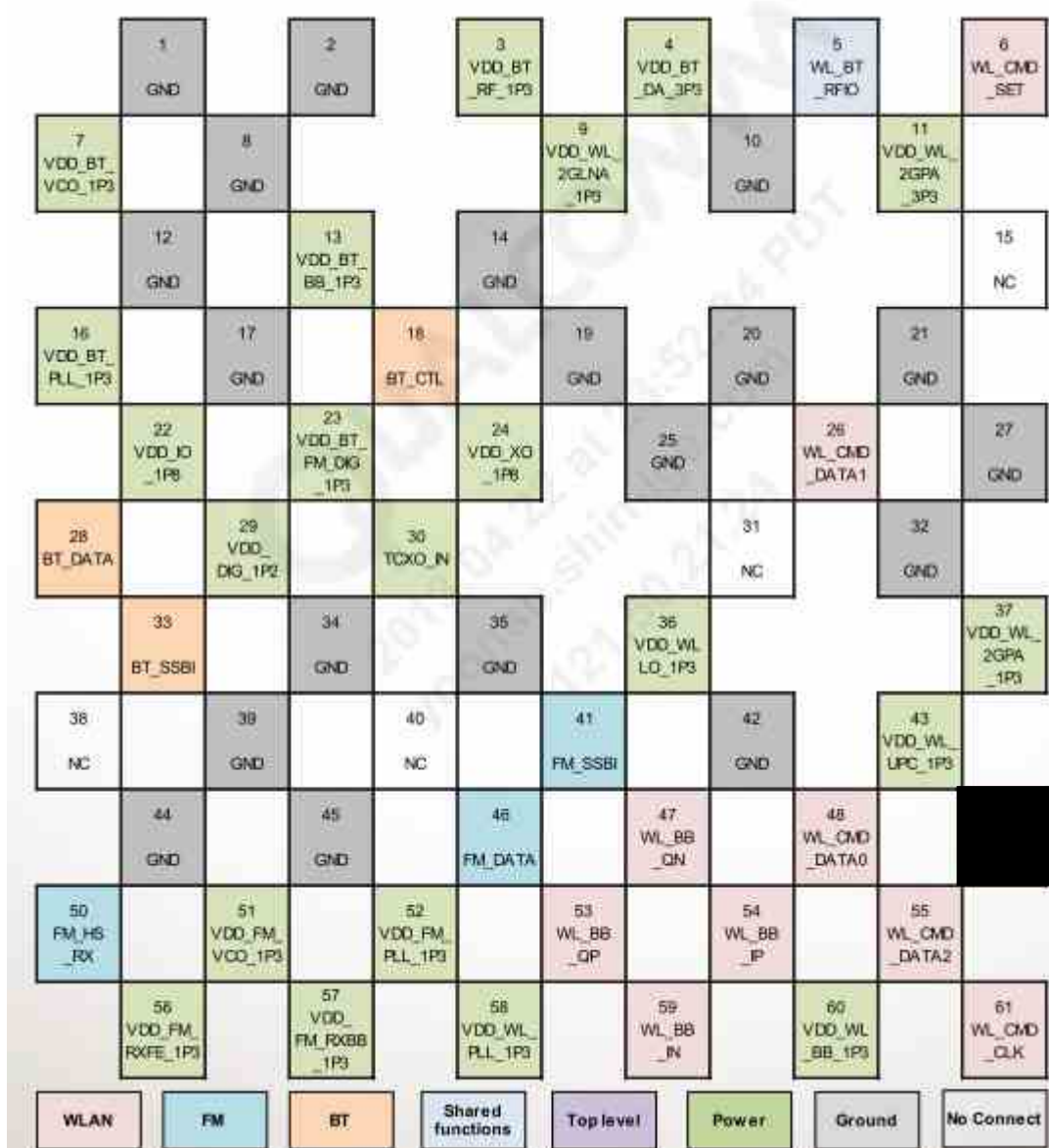
<9-8-3_Flash_LED> Rev_0.3

LED MODULE (LOW COST)



7. BGA PIN MAP

U5100- BT / Wi-Fi / FM module <TOP View>



■ Not Used

□ Used



7. BGA PIN MAP

U2100-MSM8926 (Top View) – Lower Left quadrant

AA		NC				TX_DAC0_REF		GND		VDD_MEM		VDD_MEM		VDD_MEM		VDD_CORE		VDD_CORE		GND
AB	VDD_DDR1P8		TX_DAC0_IP		TX_DAC0_W		GND		GND		GND		GND		GND		GND		GND	
AC		VREF_CA				TX_DAC1_QP		TX_DAC1_VREF		VDD_PLL0		GND		VDD_CORE		VDD_CORE		VDD_CORE		GND
AD	EBD_Z01		TX_DAC1_W		TX_DAC1_IP		TX_DAC1_QM		VDD_A1		GND		VDD_CORE		VDD_CORE		GND		VDD_MEM	
AE					GND		GND		TX_DAC1_REF		NC		GND		VDD_CORE		VDD_CORE		VDD_CORE	
AF							GND		VDD_A2		GND		GND		GND		GND		VDD_MEM	
AG		BBRX_CH0_W		BBRX_CH0_QM		BBRX_CH0_QP		VDD_A1		GND		GND		GND		GND		VDD_CORE		GND
AH	BBRX_CH0_IP		BBRX_CH0_QP		BBRX_CH0_QM		VDD_A1		GND		VDD_MEM		VDD_MEM		VDD_MEM		VDD_MEM		VDD_MEM	
AJ		VDD_P1			BBRX_CH2_IP		VDD_A2		VDD_EB0_C00		GND		VDD_CORE		GND		GND		VDD_CORE	
AK					GND		VDD_A1		GND		VDD_CORE		VDD_CORE		VDD_CORE		GND		VDD_CORE	
AL					GND		VDD_P3				GND		VDD_CORE		VDD_CORE		VDD_MPH_DSE_IP1		GND	
AM	GND		GNSS_BB_IM		GND		GND		EBD_CAL_REXT		GND		VDD_P3		NC		GND		VDD_MPH_DSE_IP1	
AN		VDD_P1		GNSS_BB_IP		GPIO_B1		GND		EBD_VREF_CA		GND				VDD_P3		GND		GND
AP			GNSS_BB_QM		GND		GND		GND							GND		MPD_DSE_INT_N		
AQ		GPIO_B9		GNSS_BB_QP		GPIO_W				GPIO_BE		GPIO_B1		NC				GND		MPD_DSE_INT_P
AT	GND					GPIO_B3		GND								GND		MPD_DSE_CLK_P		
AU		VDD_DDR1P2				GPIO_B3		GND				GPIO_F7		GND				GND		GND
AV						GPIO_B7				GND				GND		GND		MPD_DSE_CLK_N		GND
AW		GND		GND				GND				GND					VDD_DDR1P2		GND	
AX	GND		GND			GPIO_B8		VDD_DDR1P8						VDD_P1				VDD_MPH_DSE_IP2		EBD_Z01
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

 Not Used
 Used

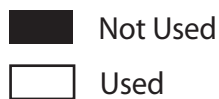
7. BGA PIN MAP

U2100-MSM8926 (Top View) - Lower Right quadrant

GND		VDD_CORE		VDD_CORE		GND		VDD_MEM		VDD_CORE		GND		VDD_P3			NC		GND	AA	
	VDD_CORE		VDD_CORE		GND		VDD_CORE		VDD_CORE		GND		GND		GPIO_27		GPIO_10		VDD_DDR_1P8	AB	
GND		GND		VDD_CORE		GND		VDD_MEM		VDD_CORE		GND		GPIO_11		GPIO_26		GPIO_25		GPIO_48	AC
	VDD_CORE		VDD_MEM		GND		GND		GND		VDD_EBI_CDC		GPIO_9				GPIO_28		GPIO_27	AD	
GND		GND		VDD_CORE		VDD_PLL1		VDD_MEM		VDD_CORE		VDD_CORE		GPIO_8			GND		GND	GPIO_24	AE
	VDD_CORE		VDD_MEM		GND		GND		GND		VDD_MEM		EBI_VREF_D0						VDD_P1	AF	
GND		GND		GND		VDD_CORE		VDD_MEM		GND		GND		VDD_P3		GND		GND		GPIO_30	AG
	VDD_CORE		VDD_MEM		GND		GND		GND		VDD_MEM		GND		GPIO_32		GND		GPIO_29	AH	
VDD_CORE		GND		VDD_MEM		VDD_CORE		VDD_EBI_CDC		VDD_CORE		GND		GND		GPIO_33		GND			AI
	VDD_CORE		GND		GND		GND		GND		GND		VDD_P1		GPIO_35		GND		GND	AK	
VDD_CORE		GND		GND		VDD_CORE		GND		VDD_CORE		GND		GPIO_38		GND		GND		GND	AL
	VDD_CORE		MPI_CS0_LDO		GND		GND		EBI_VREF_D1		GND		GND		GPIO_36		GND		VDD_P1	AM	
GND		GND		MPI_CS0_LDO_P		GND		GND		NC		GND		TD0				GND		GND	AN
	GND			MPI_CS0_LDO_N		VDD_MPI_C0		GND		NC		GND		TRST_N		GPIO_17		GPIO_22		GND	AP
MPI_CS0_LDO_P							MPI_CS0_LDO_P		GND		GND		TD0					GPIO_23		GPIO_27	AR
	MPI_CS0_LDO_N		GND		GND		MPI_CS0_LDO_P		GND		GND		TRST_N		NC			GPIO_30		VDD_P1	AT
GPIO				GND		MPI_CS0_LDO_N		GND		GND		MODE_3		NC		VDD_DDR_1P2				GPIO_21	AU
	MPI_CS0_LDO_P			GND		GND		MPI_CS0_LDO_N		GND		MODE_3		NC		NC				GPIO_18	AV
GND					GND		MPI_CS0_LDO_P		MPI_CS0_LDO_P		GND		VDD_P1		TD0		GPIO_19		GND		AW
	VDD_P1					MPI_CS0_LDO_N		MPI_CS0_LDO_N		GND		GND		VDD_DDR_1P8		TD0		GPIO_18		GND	AY

Not Used

Used



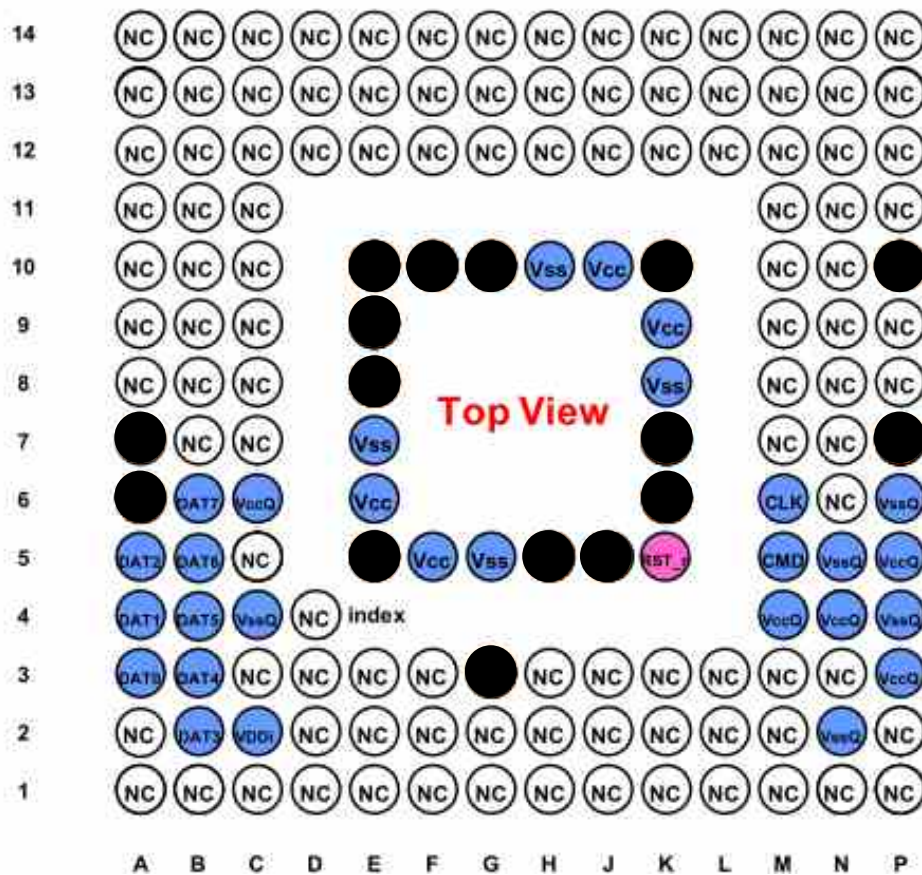
U2100-MSM8926 (Top View) - Upper Left quadrant

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A		GND		GPIO_58		GND						GND		GND		GND		VDD_DDR_1P8		GND
B	GND		GPIO_61		VDD_DDR_1P8		GPIO_59				GND		GND		GND		GND		VDD_DDR_1P8	
C		GPIO_113		GPIO_60		GPIO_57		GND				GND		GND		GND		VDD_P3		GND
D	SDC1_DATA_5						GND		GND		GND		GND		GND		GND		VREF_UW	
E		SDC1_DATA_1		SDC1_DATA_8				NC		VDD_P5		GND		GND		GND		VDD_P6		GND
F	SDC1_CLK		SDC1_DATA_5						GND		GND		GND		GND		GND		VDD_USB_1P8	
G		SDC1_DATA_11		SDC1_DATA_3		SDC1_CMD		GND		GND		VDD_APC		VDD_APC		VDD_APC		GND		USB_HST_EVSCLK
H	GND		GND		VDD_DDR_1P2		VDD_SDC_CDC		GND		GND		GND		GND		GND		GND	
J		SDC1_DATA_3		NC		GND		VDD_P7		VDD_APC		VDD_APC		GND		VDD_APC		VDD_APC		VDD_CORE
K	SDC1_DATA_7		NC		GND		VDD_SDC_CDC		GND		GND		GND		GND		GND		GND	
L		SDC2_DATA_1		SDC2_CMD		CX0		VDD_APC		VDD_APC		VDD_APC		VDD_MEM		VDD_APC		VDD_APC		VDD_CORE
M	SDC2_DATA_3		SDC2_DATA_5		SDC2_DATA_3		GND		GND		GND		GND		GND		GND		GND	
N		GND		VREF_SDC		GND		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_PLL2
P	SDC2_DATA_3		GND		NC		GND		GND		GND		GND		GND		GND		GND	
Q				GND		VDD_P5		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_MEM
T	GPIO_52				GPIO_115				GND		GND		GND		GND		GND		GND	
U		GPIO_112		GPIO_111		VREF_APC_MPM		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_APC		VDD_CORE
V			GND		TX_DACS_VREF		VDD_P3		GND		GND		GND		GND		GND		GND	
W		TX_DACS_OP						VDD_APC		VDD_APC		VDD_MEM		VDD_CORE		VDD_CORE		VDD_MEM		VDD_MEM
Y	TX_DACS_GM				GND		VDD_AE		GND		GND		GND		GND		GND		GND	

Not Used

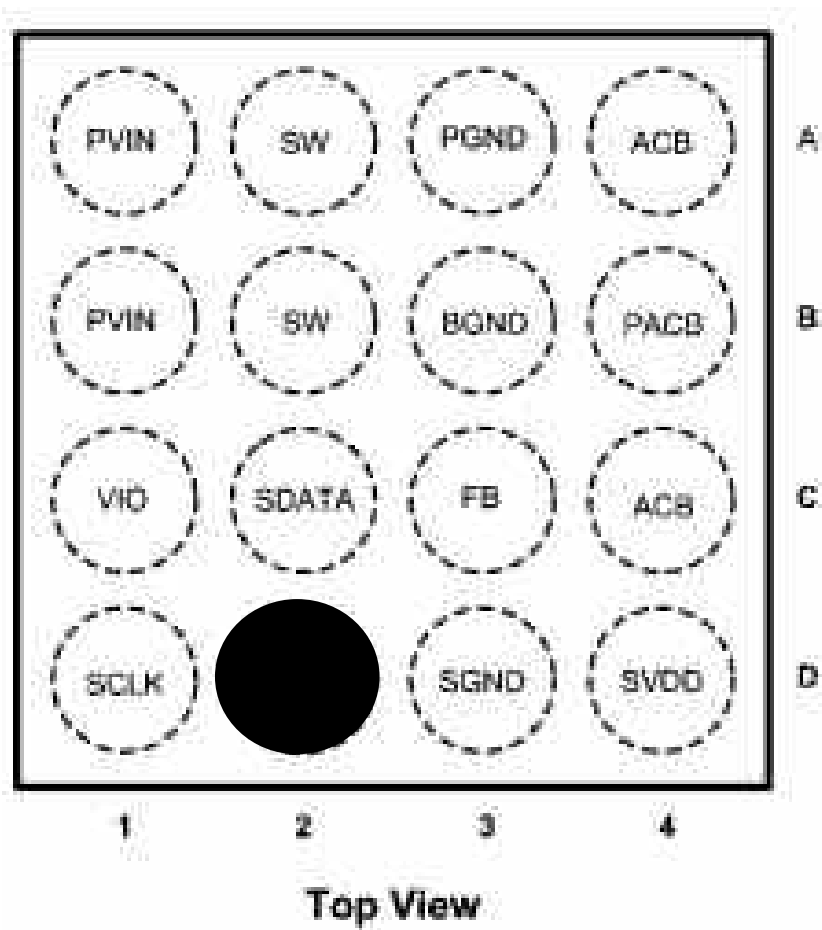
Used

U3200- eMMC (Top View)



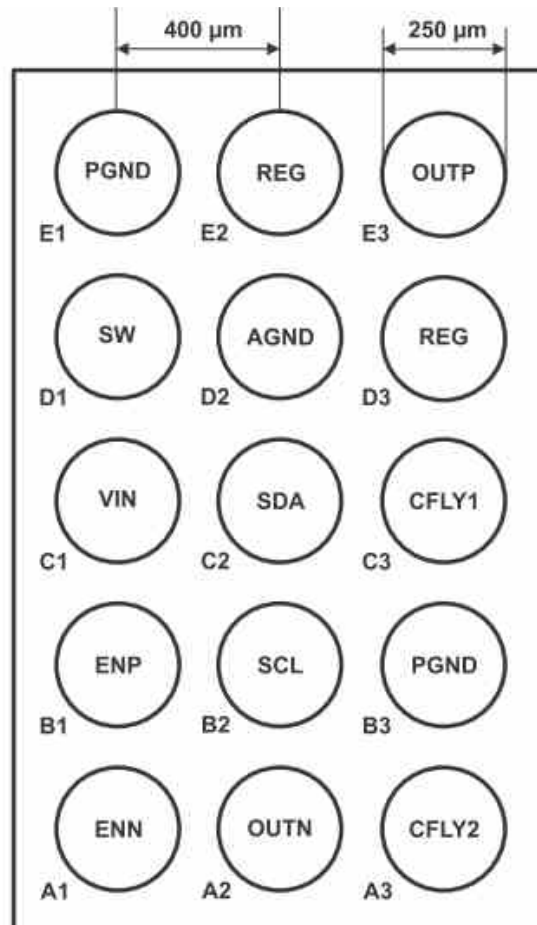
● Not Used

U1400 –IC, DC,DC Converter (Top View)



● Not Used

U9301 –IC, DC,DC Converter (Bottom View)



PIN		I/O	DESCRIPTION
NAME	NO.		
ENN	A1	I	Enable pin for V_{NEG} rail
OUTN	A2	O	Output pin of the negative charge pump (V_{NEG})
CFLY2	A3	I/O	Negative charge pump flying capacitor pin
ENP	B1	I	Enable pin for V_{POS} rail
SCL	B2	I/O	I ² C interface clock signal pin
PGND	B3, E1	G	Power ground
VIN	C1	I	Input voltage supply pin
SDA	C2	I/O	I ² C interface data signal pin
CFLY1	C3	I/O	Negative charge pump flying capacitor pin
SW	D1	I/O	Switch pin of the boost converter
AGND	D2	G	Analog ground
REG	D3, E2	I/O	Boost converter output pin
OUTP	E3	O	Output pin of the LDO (V_{POS})

ALL USED

7. BGA PIN MAP

U4100 –IC,PMIC (Top View)

1 VDD_ GR2	2 VREG_ L3	3 VREG_ L2	4 VDD_ GR1		5 XTAL_ 19M_IN	6 XTAL_ 19M_OUT		7 VDD_ GR6	8 VSW_ S1	9 VDD_ S1	10 VDD_ S1
11 VDD_ GR2	12 VREG_ L24	13 RF_CLK1	14 VREG_ L13	15 VREG_ L1	16 VREG_ L11	17 GND_ XO	18 VREG_ L17	19 VREG_ L16	20 VREG_ S1	21 VSW_ S1	22 GND_ S1
23 GND_ S3	24 VREG_ L26	25 VREG_ L5	26 NC	27 VDD_ GR4	28 NC	29 VREG_ L19	30 VREG_ L17	31 BB_CLK1	32 VREG_ L18	33 GND_ S1	34 VDD_ S2
35 VSW_ S3	36 VREG_ S3	37 VREG_ L4	38 VREG_ L10	39 RF_CLK2	40 NC	41 VDD_ GR7	42 BB_CLK2	43 VREG_ L15	44 BB_CLK1 _EN	45 VSW_ S2	46 VSW_ S2
47 VDD_ S3	48 VDD_ L25	49 VREG_ L14	50 VREG_ L28	51 VREG_ L23	52 VREG_ L22	53 VREG_ L21	54 NC	55 VREG_ L20	56 VREG_ S2	57 GND_ S2	58 VSW_ S2
	59 VDD_ GR5	60 VREG_ GR5	61 VREF_ LPDDR	62 GND_ XO_BUFF	63 GND_ XOADC	64 GND	65 GND_ DRV	66 VIB_ DRV_N	67 GND_ S2	68 GND_ S2	
69 GND_ S4	70 VREG_ L25	71 VREG_ L12	72 AMUX_1	73 AMUX_2	74 GND	75 GND	76 VCOIN	77 PON_ RESET_N	78 VREG_ S5	79 VDD_ S5	80 VSW_ S5
81 VSW_ S4	82 NC	83 DNC	84 DNC	85 CBL_ PWR_N	86 GND	87 GND	88 SLEEP_ CLK1	89 NC	90 NC	91 NC	92 GND_ S5
93 VDD_ S4	94 VOUT_ LVS1	95 VREG_ S4	96 BMS_ CSN	97 USB_ ID_IN	98 GND	99 GND	100 PS_HOLD	101 RESIN_ IN	102 KPD_ PWR_N	103 VLED_ FLASH	104 VDD_ FLASH
	105 VREG_ L6	106 VREG_ L6	107 BMS_ CSP	108 OPT_2	109 GND	110 GND_ OVP	111 GND	112 SPMI_ CLK	113 GND_ WLEDI	114 GND_ WLEDI	
115 VDD_ GR3	116 VREG_ L7	117 GND_ REF	118 VREG_ L8	119 OPT_1	120 XO_ THERM	121 AVDD_ BYP		123 SPMI_ DATA	124 GPIO_1	125 WLED_1	126 VDD_ WLED
127 VREG_ L27	128 VREG_ L9	129 REF_ BYP	130 MPP_7	131 BMS_ BYP	132 MPP_5	133 NC	134 GPIO_7	135 GPIO_4	136 GPIO_2	137 OVP_ WLED	138 VSW_ WLED
139 VBAT	140 BATFET_ CP_DRV		142 GND_ CHG	143 VCHG		145 NC	146 MPP_2	147 NC	148 GPIO_5	149 GPIO_3	150 GND_ WLED
151 VBAT	152 VPH_ PWR	153 GND_ CHG_HP	154 VSW_ CHG	155 VCHG	156 VBAT_ SNS		158 MPP_3	159 MPP_1	160 GPIO_6	161 USB_IN	162 OVP_ OUT
163 VBAT	164 VPH_ PWR	165 GND_ CHG_HP	166 VSW_ CHG		167 VDRV_P			169 VREF_ BAT_THM	170 VPRE_ BYP	171 OVP_CP_ DRV	172 USB_IN

Input power management	Output power management	General housekeeping	User interfaces	IC-level interfaces	MPPs and GPIOs	No connection	Power	Ground
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Not Used

Used

U2003 –WTR1605L (Top View) Lower Left quadrant

74 VDD_RF1 _G_VCO		75 GND		76 DRX_ BB_IP		77 GND		78 VDD_RF1 _S_PLL	
	84 GND		85 GNSS_ BB_QM		86 DRX_ BB_IM		87 GND		88 GND
93 VDD_RF1 _G_PLL		94 GND		95 SSBI_ PRX_DRX		96 GND		97 PRX_ BB_QM	
	103 VDD_DIO		104 GP_ DATA1		105 SSBI_ TX_GNSS		106 GND		107 PRX_ BB_QP
112 GND		113 GND		114 VDD_RF2 _T_PLL		115 VDD_RF1 _T_SYN		116 VDD_RF2 _FBRX	
	122 GND		123 GP_ DATA0		124 GND		125 GND		126 VDD_RF2 _T_BB
131 XO_IN		132 GND		133 GND				134 GND	
	140 GND		141 GND				142 GND		143 GND
147 VDD_RF2 _XO		148 GND		149 VDD_RF1 _T_VCO				150 GND	
	156 GND		157 VDD_RF2 _T_VCO		158 GND		159 GND		160 TX_ BB_IM

☐ Used

U2003 –WTR1605L (Top View) Lower Right quadrant

	79 VDD_RF1 _P_PLL		80 VDD_RF2 _P_VCO		81 GND		82 GND		83 GND
88 GND		89 GND		90 VDD_RF1 _P_VCO					
	98 VDD_RF2 _P_BB		99 PRX_ BB_IP		100 VDD_RF2 _P_RX		101 GND		
107 PRX_ BB_QP		108 PRX_ BB_IM				110 GND		111 GND	
	117 PDET_ RFFB		118 GND		119 GND		120 GND		121 TX_ HB2_OUT
126 VDD_RF2 _T_BB		127 DAC_ REF		128 GND		129 VDD_RF2 _T_DA			
	135 VDD_RF1 _T_LO		136 VDD_RF1 _T_UPC		137 VDD_RF1 _T_DA				139 TX_ MB3_OUT
143 GND				144 GND		145 GND			
	151 TX_ BB_IP		152 TX_ BB_QP						
160 TX_ BB_IM		161 TX_ BB_QM		162 TX_ LB1_OUT				164 GND	

Not Used

Used

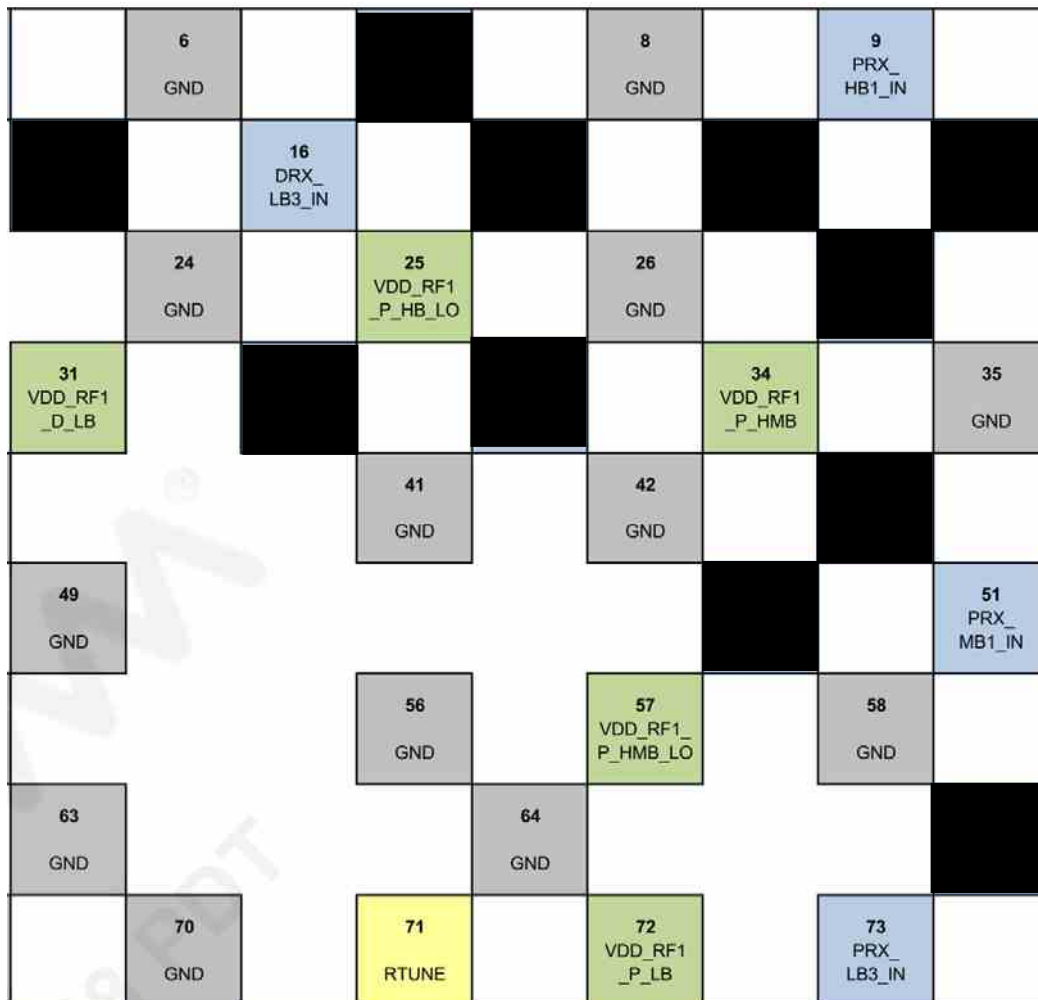
U2003 –WTR1605L (Top View) Upper Left quadrant

					3 GND		4 DRX_ HB1_IN		
10 GND		11 VDD_RF1 _D_MB						14 VDD_RF1 _D_LB_LO	
			21 GND		22 VDD_RF1 _D_HB		23 GND		
28 DRX_ MB1_IN									
	36 GNSS_ RF_INP		37 DNC		38 VDD_RF1 _D_LOM		39 GND		40 GND
44 GNSS_ RF_INM		45 GND		46 GND		47 GND		48 VDD_RF1 _S_VCO	
	52 VDD_RF1_G _LNA		53 GNSS_ BB_IM		54 VDD_RF2 _D_BB				55 GND
59 VDD_RF1 _G_BB		60 GNSS_ BB_IP		61 DRX_ BB_QP				62 VDD_RF2 _S_VCO	
	66 GND		67 GNSS_ BB_QP		68 DRX_ BB_QM				69 GND

■ Not Used

□ Used

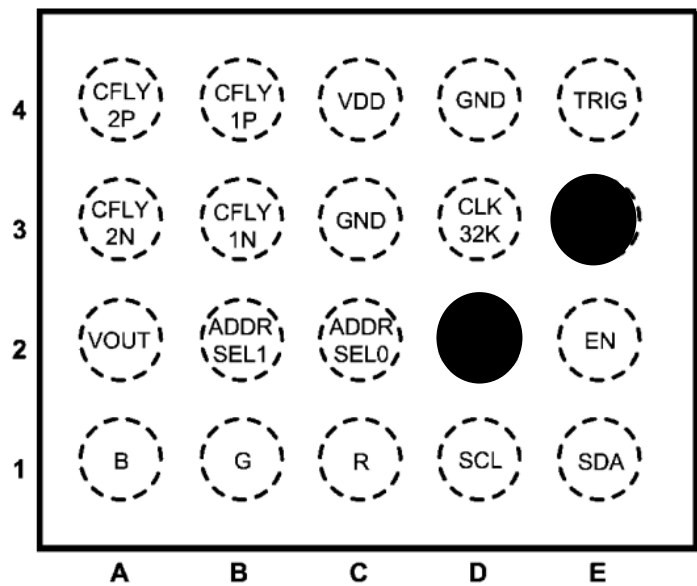
U2003-WTR1605L (Top Viw) Upper Right quadrant



Not Used

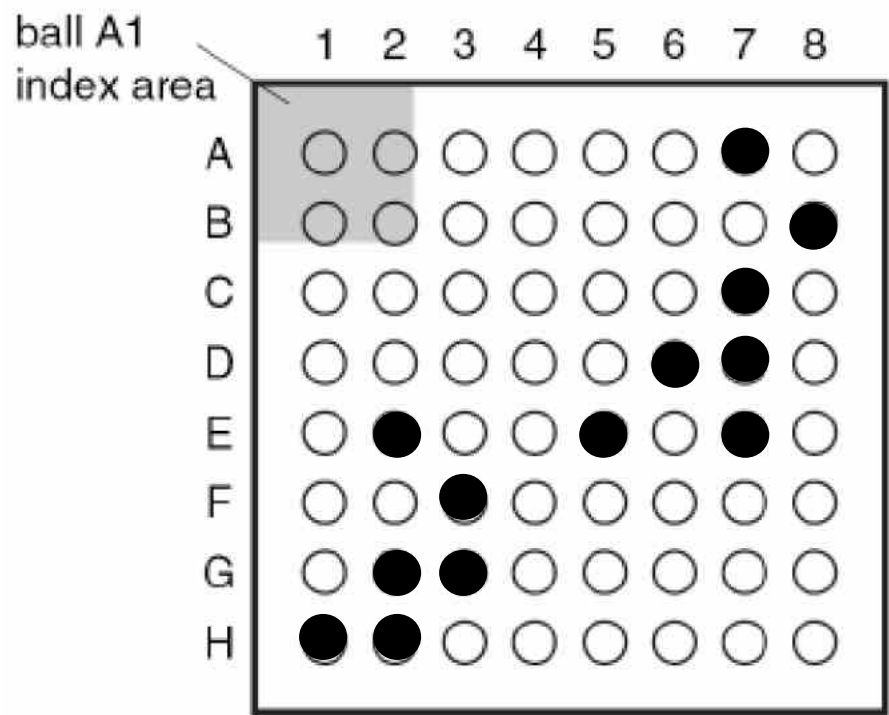
Used

U4600 – IC,LED Driver (Top View)



Not Used

U5200– IC,NFC (Top View)



● Not Used